

voice vignette

version 0.5.6

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0. Basic installation

```
# Development version from GitHub
install.packages('devtools')
devtools::install_github('filipezabala/voice')

# Stable version from CRAN
install.packages('voice')
```

0.1 For Windows Users

If you're compiling R packages from source, you may need to install [RTools](#), a collection of Windows-specific build tools for R.

0.2 For macOS Users

If you're compiling packages, ensure you have [Xcode Command Line Tools](#) installed. You also may need [macOS tools](#).

```
# Install Xcode on MacOS
xcode-select --install
```

More details may be found at <https://github.com/filipezabala/voice>.

1. Extract features

1.1 Load packages and audio files

```
# packs
library(voice)
library(tidyverse)

# get path to audio file
wavDir <- list.files(system.file('extdata', package = 'wrassp'),
                     pattern = glob2rx('*.wav'), full.names = TRUE)
```

1.2 Extract features

```
# minimal usage
M <- voice::extract_features(wavDir)
glimpse(M)
#> Rows: 1,196
#> Columns: 13
#> $ section_seq      <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16...
#> $ section_seq_file <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16...
#> $ wav_path         <chr> "/Library/Frameworks/R.framework/Versions/4.6/Resourc...
#> $ f0               <dbl> NA, NA, NA, NA, NA, NA, NA, 115.8593, 108.9439, 107.4...
#> $ f1               <int> NA, NA, NA, NA, 185, 260, 254, 277, 261, 231, 177, 19...
#> $ f2               <int> 1854, 1886, 1749, 1888, 1962, 1973, 2026, 2037, 2130,...
#> $ f3               <int> NA, 2893, 2676, 2659, 2639, 2676, 2993, 2932, 3016, 2...
#> $ f4               <int> 3113, 3708, 3509, 3658, 3248, 3239, 3830, 3479, 3561,...
#> $ f5               <int> 4191, 4678, 4502, 4331, 3653, 3836, 4602, 4585, 4720,...
#> $ f6               <int> 5226, 5659, 5035, 5177, 5208, 5146, 5233, 5390, 5366,...
#> $ f7               <int> 6077, 6725, 6526, 6518, 6493, 6567, 6603, 6532, 6510,...
#> $ f8               <int> 6675, NA, NA, NA, 7681, 7751, 7803, NA, 7835, 7614, 7...
#> $ gain             <dbl> 21.63347, 22.76034, 28.52825, 29.67069, 36.25124, 43...
```

2. Tag

```
# creating Extended synthetic data
E <- dplyr::tibble(subject_id = c(1,1,1,2,2,2,3,3,3), wav_path = wavDir)
E
#> # A tibble: 9 × 2
#>   subject_id wav_path
#>   <dbl> <chr>
#> 1         1 /Library/Frameworks/R.framework/Versions/4.6/Resources/library/wra...
#> 2         1 /Library/Frameworks/R.framework/Versions/4.6/Resources/library/wra...
#> 3         1 /Library/Frameworks/R.framework/Versions/4.6/Resources/library/wra...
#> 4         2 /Library/Frameworks/R.framework/Versions/4.6/Resources/library/wra...
#> 5         2 /Library/Frameworks/R.framework/Versions/4.6/Resources/library/wra...
#> 6         2 /Library/Frameworks/R.framework/Versions/4.6/Resources/library/wra...
```

```
#> 7      3 /Library/Frameworks/R.framework/Versions/4.6/Resources/library/wra...
#> 8      3 /Library/Frameworks/R.framework/Versions/4.6/Resources/library/wra...
#> 9      3 /Library/Frameworks/R.framework/Versions/4.6/Resources/library/wra...

# minimal usage
voice::tag(E)
#> # A tibble: 9 × 7
#>   wav_path    f0_tag_mean f0_tag_sd f0_tag_vc f0_tag_median f0_tag_iqr f0_tag_mad
#>   <chr>          <dbl>    <dbl>    <dbl>         <dbl>    <dbl>    <dbl>
#> 1 /Library/...    85.4     17.6     0.206         76.1     29.4     7.53
#> 2 /Library/...    85.4     15.6     0.183         80.1     27.8     14.4
#> 3 /Library/...    84.6     13.0     0.154         78.8     23.9     14.0
#> 4 /Library/...    84.8     14.5     0.171         79.1     28.1     11.9
#> 5 /Library/...    86.0     14.7     0.170         78.7     30.0     11.0
#> 6 /Library/...    82.9     15.6     0.188         74.8     23.8     4.78
#> 7 /Library/...    78.2     16.2     0.207         73.5     13.4     6.82
#> 8 /Library/...    84.5     14.5     0.172         78.1     17.8     8.95
#> 9 /Library/...    81.0     12.2     0.151         75.9     23.1     9.14

# canonical data
voice::tag(E, groupBy = 'subject_id')
#> # A tibble: 3 × 7
#>   subject_id f0_tag_mean f0_tag_sd f0_tag_vc f0_tag_median f0_tag_iqr f0_tag_mad
#>   <dbl>          <dbl>    <dbl>    <dbl>         <dbl>    <dbl>    <dbl>
#> 1           1    85.1     15.3     0.180         78.3     26.8     11.9
#> 2           2    84.6     14.9     0.176         76.4     28.3     7.97
#> 3           3    81.0     14.6     0.180         75.6     21.6     8.68
```

3. Visualization

3.1 Get audio

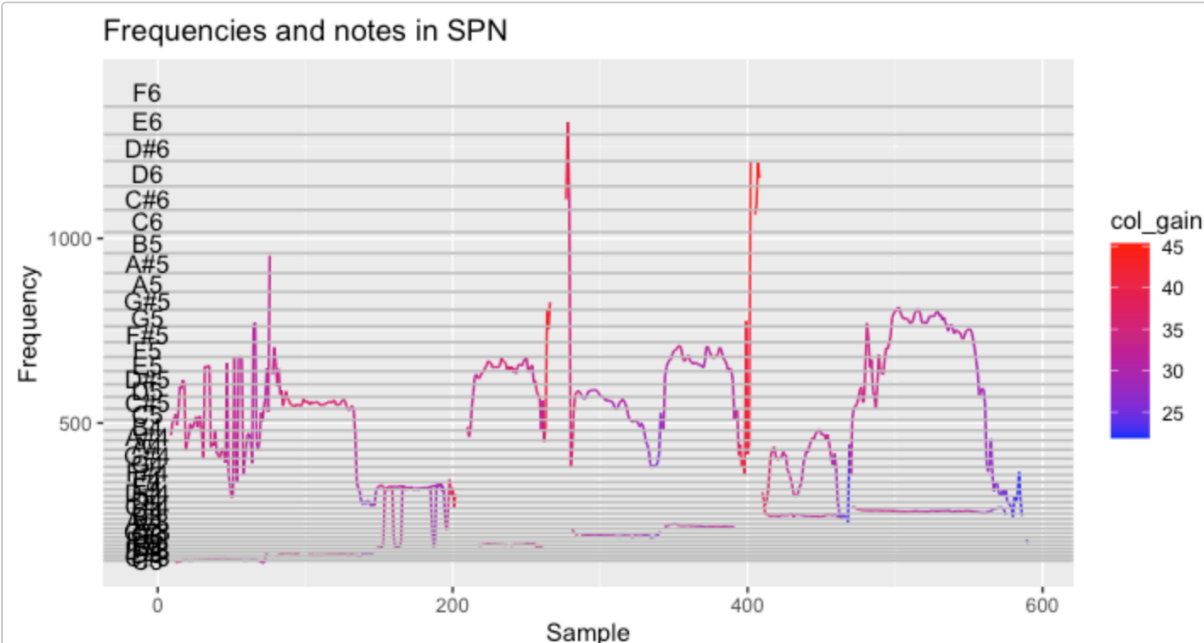
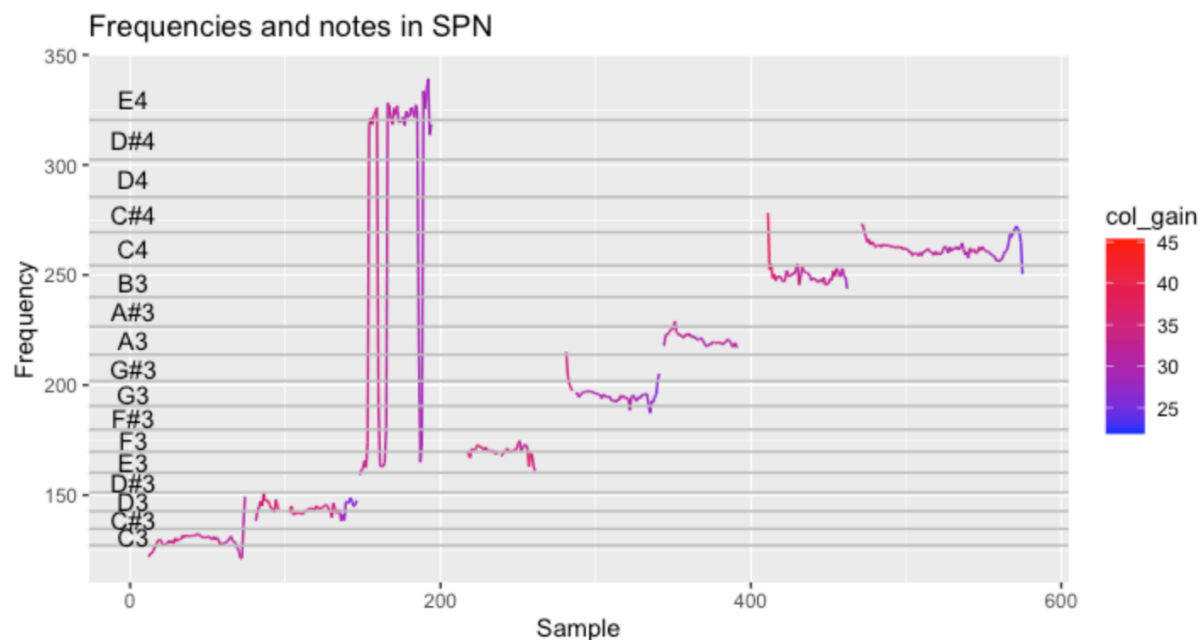
```
url0 <- 'https://github.com/filipezabala/voiceAudios/raw/refs/heads/main/wav/doremi.wav'
download.file(url0, paste0(tempdir(), '/doremi.wav'), mode = 'wb')
voice::embed_audio(url0) # See https://github.com/mccarthy-m-g/embedr for more details.
```

3.2 Media data

```
M <- voice::extract_features(tempdir())
summary(M)
```

3.3 Plot

```
voice::piano_plot(M, 0) # f0
voice::piano_plot(M, 0:1) # f0 + f1
```



3.4 Assign notes

```
(f0_spn <- voice::assign_notes(M, fmt = 0, min_points = 22, min_percentile = .85)) # f0
(f1_spn <- voice::assign_notes(M, fmt = 1, min_points = 22, min_percentile = .85)) # f1
```

3.5 Sheet music

Must have [MuseScore](#) and [gm](#).

3.5.1 Notes sequence of f0

```
library(gm)
line_0 <- gm::Line(as.character(f0_spn))
```

```
m0 <- gm::Music() +
  gm::Meter(4, 4) +
  line_0
gm::show(m0, to = c('score', 'audio'))
```



3.5.2 Notes sequences of f0 and f1

```
line_0 <- gm::Line(as.character(f0_spn))
line_1 <- gm::Line(as.character(f1_spn))
m1 <- gm::Music() +
  gm::Meter(4, 4) +
  line_0 + line_1
gm::show(m1, to = c('score', 'audio'))
```



4. Advanced installation

Python-based functions `diarize` and `extract_features` (when the latter is inferring `f0_praat` and `fmt_praat` features) require a configured Python environment.

4.1 Ubuntu

The following steps are used to fully configure `voice` on Ubuntu 24.04 LTS (Noble Numbat). Reports of inconsistencies are welcome.

4.1.1. [Curl](#)

Command line tool and library for transferring data with URLs.

```
# installing dependencies
sudo apt-get update
sudo apt-get install -y libssl-dev autoconf libtool make
# installing curl
sudo apt install curl
# verify installation
curl --version
```

4.1.2. [ffmpeg](#)

ffmpeg is a cross-platform solution to record, convert and stream audio and video.

```
sudo apt-get update
sudo apt-get install ffmpeg
```

4.1.3. Audio drivers and extra packages

```
sudo apt-get update
sudo apt-get install portaudio19-dev libasound2-dev libfontconfig1-dev libmagick++-dev
libxml2-dev libharfbuzz-dev libfribidi-dev libgdal-dev cmake cmake-doc ninja-
build
```

4.1.4. [MuseScore](#)

MuseScore is an open source notation software.

```
sudo add-apt-repository ppa:mscore-ubuntu/mscore-stable
sudo apt-get update
sudo apt-get install musescore
```

4.1.5. [R](#)

R is a free software environment for statistical computing and graphics. To find out your Ubuntu distribution use `lsb_release -a` at terminal.

```
sudo sh -c 'echo "deb https://cloud.r-project.org/bin/linux/ubuntu focal-cran40/" >>
/etc/apt/sources.list.d/cran.list'
sudo apt-key adv --keyserver keyserver.ubuntu.com --recv-keys E084DAB9
sudo apt-key adv --keyserver keyserver.ubuntu.com --recv-keys 51716619E084DAB9
gpg --a --export E084DAB9 | sudo apt-key add -

sudo add-apt-repository ppa:c2d4u.team/c2d4u4.0+

sudo apt-get update && sudo apt-get upgrade
sudo apt-get install r-base r-base-dev
```

4.1.6. [RStudio](#)

RStudio is an Integrated Development Environment (IDE) for R. Check for updates [here](#).

```
sudo apt-get update
sudo apt-get install gdebi-core
wget https://download1.rstudio.org/electron/jammy/amd64/rstudio-2025.05.0-496-amd64.deb
sudo gdebi rstudio-2025.05.0-496-amd64.deb
```

4.1.9. R packages

“Packages are the fundamental units of reproducible R code.” [Hadley Wickham and Jennifer Bryan](#). The installation may take several minutes. At terminal run:

```
sudo R
```

Running R as super user paste the following, row by row:

```
packs <- c('audio','reticulate','R.utils','seewave','tidyverse','tuneR','wrassp')
install.packages(packs, dep = TRUE)
update.packages(ask = FALSE)
devtools::install_github('egenn/music')
devtools::install_github('flujoo/gm')
```

To configure the `gm` package.

```
usethis::edit_r_environ()
```

Add the line `MUSESCORE_PATH=/usr/bin/mscore` to `/root/.Renviro` file. To exit use `:wq` at VI. Save and restart the R/RStudio session.

4.1.10. [Miniconda](#)

Miniconda is a free minimal installer for [conda](#), an open source package, dependency and environment management system for any language—Python, R, Ruby, Lua, Scala, Java, JavaScript, C/ C++, FORTRAN and more, that runs on Windows, macOS and Linux.

Follow the instructions at <https://docs.conda.io/en/latest/miniconda.html>.

At terminal:

```
cd ~/Downloads/
wget -r -np -k https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh
cd repo.anaconda.com/miniconda/
bash Miniconda3-latest-Linux-x86_64.sh
```

Do you accept the license terms? [yes|no] yes.

Miniconda3 will now be installed into this location: `/home/user/miniconda3` [ENTER]

You can undo this by running `conda init --reverse $SHELL?` yes

Do you wish the installer to initialize Miniconda3 by running `conda init?` yes.

Close and reopen terminal.

```
conda update -n base -c defaults conda
```

The following packages will be INSTALLED/REMOVED/UPDATED/DOWNGRADED:... Proceed ([y]/n)? y

```
conda create -n pyvoice python=3.12
```

The following (NEW) packages will be downloaded/INSTALLED:... Proceed ([y]/n)? y

```
conda activate pyvoice
pip install -r
https://raw.githubusercontent.com/filipezabala/voice/master/requirements.txt
```

4.2 MacOS

The following steps are used to fully configure voice on [MacOS Sonoma and Tahoe](#). Reports of inconsistencies are welcome.

4.2.1. [Homebrew](#)

Install Homebrew, 'The Missing Package Manager for macOS (or Linux)' and remember to brew doctor eventually. At terminal (command + space 'terminal') run:

```
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

4.2.2. [wget](#)

GNU Wget is a free software package for retrieving files using HTTP, HTTPS, FTP and FTPS, the most widely used Internet protocols. It is a non-interactive commandline tool, so it may easily be called from scripts, cron jobs, terminals without X-Windows support, etc.

```
brew install wget
```

4.2.3. [Python](#)

Python is a programming language that integrate systems. According to [this](#) post, it is recommended to install Python 3.8 and 3.9 and make it consistent.

```
brew install python@3.12
python3 --version
pip3 --version
```

4.2.4. [ffmpeg](#)

ffmpeg is a cross-platform solution to record, convert and stream audio and video. The installation may take several minutes.

```
brew install ffmpeg
```

4.2.5. [XQuartz](#)

The XQuartz project is an open-source effort to develop a version of the [X.Org X Window System](#) that runs on macOS.

- Download and run <https://github.com/XQuartz/XQuartz/releases/download/XQuartz-2.8.5/XQuartz-2.8.5.pkg>
- Will take around 320 MB of disk space
- Send XQuartz-2.8.5.dmg to Trash

4.2.6. [MacPorts](#)

Follow the instructions from <https://guide.macports.org/chunked/installing.macports.html>.

4.2.7. [tcllib](#)

```
sudo port selfupdate && sudo port upgrade tcllib
sudo port install tcllib
```

4.2.8. [MuseScore](#)

MuseScore is an open source notation software.

- Download and run <https://musescore.org/en/download/musescore.dmg>
- Drag MuseScore 4 to Applications folder
- Will take around 320 MB of disk space
- Unmount MuseScore-4.2.1 virtual disk and send MuseScore-4.2.1.240230937.dmg to Trash

4.2.9. [R](#)

R is a free software environment for statistical computing and graphics.

- Download and run the pkg file according to you architecture from <https://cran.r-project.org/bin/macosx/>
- Will take around 180 MB of disk space

4.2.10. [RStudio](#)

RStudio is an Integrated Development Environment (IDE) for R.

- Download and run <https://download1.rstudio.org/electron/macos/RStudio-2025.05.0-496.dmg>
- Drag RStudio to Applications folder
- Will take around 770 MB of disk space
- Unmount RStudio virtual disk and send RStudio-2025.05.0-496.dmg to Trash
- Type command + space 'rstudio'
- Tools > Global Options... > Appearance > Merbivore (Restart required)

4.2.11. R packages

“Packages are the fundamental units of reproducible R code.” [Hadley Wickham and Jennifer Bryan](#). Type command + space 'terminal'

```
sudo R
```

Running R as super user paste the following, one line at a time.

```
packs <- c('audio','reticulate','R.utils','seewave','tidyverse','tuneR','wrassp')
install.packages(packs, dep = TRUE)
update.packages(ask = FALSE)
devtools::install_github('egenn/music')
devtools::install_github('flujoo/gm')
```

4.2.12. [Miniconda](#)

Miniconda is a free minimal installer for [conda](#), an open source package, dependency and environment management system for any language—Python, R, Ruby, Lua, Scala, Java, JavaScript, C/ C++, FORTRAN and more, that runs on Windows, macOS and Linux.

For 64-bit version use

```
cd ~/Downloads
wget -r -np -k https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-x86_64.sh
cd repo.anaconda.com/miniconda/
bash Miniconda3-latest-MacOSX-x86_64.sh
```

For M1 version use

```
cd ~/Downloads
wget -r -np -k https://repo.anaconda.com/miniconda/Miniconda3-latest-MacOSX-arm64.sh
cd repo.anaconda.com/miniconda/
bash Miniconda3-latest-MacOSX-arm64.sh
```

In order to continue the installation process, please review the license agreement. Please, press ENTER to continue ENTER.

You can undo this by running `conda init --reverse $SHELL? yes`

Close and reopen terminal.

```
export PATH="~/miniconda3/bin:$PATH"
conda update -n base -c defaults conda
```

The following packages will be INSTALLED/REMOVED/UPDATED/DOWNGRADED:... Proceed ([y]/n)? y

```
conda create -n pyvoice python=3.12
```

The following (NEW) packages will be downloaded/INSTALLED:... Proceed ([y]/n)? y

Close and reopen terminal.

```
conda activate base
conda activate pyvoice
pip install -r
https://raw.githubusercontent.com/filipezabala/voice/master/requirements.txt
```

5. Diarize

```
# download
url0 <- 'https://github.com/filipezabala/voiceAudios/raw/main/wav/sherlock0.wav'
wavDir <- normalizePath(tempdir())
download.file(url0, paste0(wavDir, '/sherlock0.wav'), mode = 'wb')
```

Diarization can be performed to detect speaker segments (i.e., 'who spoke when').

```
# diarize
voice::diarize(fromWav = wavDir, toRttm = wavDir, token = 'YOUR_TOKEN')
```

The `voice::diarize()` function creates Rich Transcription Time Marked (RTTM)¹ files, space-delimited text files containing one turn per line defined by NIST - National Institute of Standards and Technology. The RTTM files can be read using `voice::read_rttm()`.

```
# read_rttm
(rttm <- voice::read_rttm(wavDir))
```

Finally, the audio waves can be automatically segmented.

```
# split audio wave
voice::splitw(fromWav = wavDir, fromRttm = wavDir, to = wavDir)
dir(wavDir, pattern = '.[Ww][Aa][Vv]$')
```

-
1. See Appendix C at <https://www.nist.gov/system/files/documents/itl/iad/mig/KWS15-evalplan-v05.pdf>. ↩